UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2019

Roll No. 019599

Paper: Solid State Physics-II
Course Code: PHY-308 Part – II

Time: 2 Hrs. 45 Min. Marks: 60

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Question No.2: Answer the following short questions. **(10x2=20)**

1 Define Bloch function and Bloch theorem.

2 Explain briefly intrinsic mobility.

3 Differentiate between metals and insulators on the basis of orbital band theory.

4 What are significances of Hall-co-efficient?

5 Plot the distribution of probability 'p' in the lattice for $|\Psi +|^2, |\Psi -|^2$ *and for* pure travelling wave.

6 What is meant by density of energy states in metals?

7 Define and draw Fermi sphere.

8 What factors affect the resistivity of electrical materials?

9 State Weidman Franz law.

10 Explain the difference between direct and indirect band gap materials.

Q.3. (10)

a. Give quantitative explanation of energy gap formation under periodic potential.

b. Plot Fermi dirac distribution function for various temperatures.

Q.4. (10)

a. Show that the wavelength associated with an electron having an energy equal to Fermi energy is given by

$$\lambda_F = 2\left(\frac{\pi}{3n}\right)^{1/3}$$

b. Derive an expression for intrinsic carrier concentration in a semiconductor.

Q.5. (10)

a. Consider Free electron gas in 3-dimensions, then by using periodic boundary conditions, derive expressions for Fermi energy and density of states for the system.

b. Explain collision time and mean free path as applied to free electrons.

So the Bloch theorem is ...

So $u_k(x)$ has a periodicity of a

$$u_k(x) = u_k(x+a).$$

2019 LQ Quantitative explanation of energy gap formation under periodic potential

Lecture 7:-

KRONIG PENNEY MODEL:-

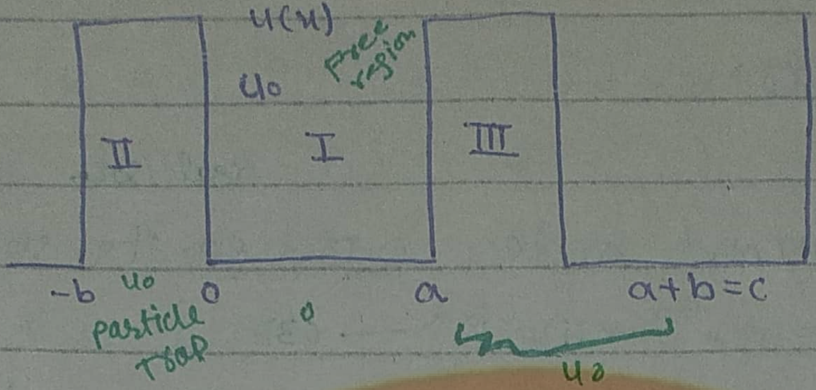
In Kronig penney model, consider the simple potential in the form of an array of square wells as shown in fig (a). So, the periodic potential of form is,

$$U(x) = \begin{cases} 0 & 0 < x < a \\ U_0 & a < x < a+b \end{cases}$$

$-b < x < 0$

The schrodinger wave eq is,

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E-U)\psi = 0 \quad \text{— (1)}$$



For Region I:-

In a region I, the P.E is zero. The particle is free so eq (1), becomes,

$$\frac{d^2\psi_1}{dx^2} + \frac{2m}{\hbar^2}E\psi_1 = 0 \qquad (\text{TDSWE})$$

Let $K^2 = \frac{2mE}{\hbar^2}$ put in above eq,

$$\frac{d^2\psi_1}{dx^2} + k^2\psi_1 = 0$$

This is second order d.E

Its solution is, plane wave soln

$$\psi_1(x) = Ae^{ikx} + Be^{-ikx} \quad \text{— (2)}$$

For Region II:-

In region II the P.E is U_0,

so eq (1) becomes,

$$\frac{d^2\psi_2}{dx^2} - \frac{2m}{\hbar^2}(U_0 - E)\psi_2 = 0$$

$$Q^2 = \frac{2m}{\hbar^2}(U_0 - E)$$

$$\begin{array}{cccc} A & B & C & D \end{array}$$
$$\begin{vmatrix} 1 & 1 & -1 & -1 \\ ik & -ik & -Q & Q \\ e^{ika} & e^{-ika} & e^{-Qb}e^{ik(a+b)} & -e^{Qb}e^{ik(a+b)} \\ ike^{ika} & -ike^{-ika} & -Qe^{-Qb}e^{i(a+b)k} & Qe^{Qb}e^{ik(a+b)} \end{vmatrix} = 0$$

By $C_1 - C_2$ and $C_3 - C_4$ in above determinant so,

$$\begin{vmatrix} 0 & 1 & 0 & -1 \\ 2ik & -ik & -2Q & Q \\ 2i\sin ka & e^{-ika} & e^{ik(a+b)}2\sinh Qb & -e^{ik(a+b)}e^{Qb} \\ 2ik\cos ka & -ike^{-ika} & -Qe^{ik(a+b)}2\cosh Qb & e^{ik(a+b)}e^{Qb} \end{vmatrix}$$

By $C_4 + C_2$, $\frac{1}{2i}C_1$ and $-\frac{1}{2}C_3$

$$\begin{vmatrix} 0 & 1 & 0 & 0 \\ k & -ik & Q & Q-ik \\ \sin ka & e^{-ika} & -e^{ik(a+b)}\sinh Qb & -e^{ik(a+b)}e^{Qb}+e^{ika} \\ k\cos ka & -ike^{-ika} & Qe^{ik(a+b)}\cosh Qb & e^{ik(a+b)}e^{Qb}-ike^{ika} \end{vmatrix}$$

By eliminating 1^{st} row and 2^{nd} column.

$$(-)\begin{vmatrix} k & Q & Q-ik \\ \sin ka & -e^{ik(a+b)}\sinh Qb & -e^{ik(a+b)}e^{Qb}+e^{ika} \\ k\cos ka & Qe^{ik(a+b)}\cosh Qb & Qe^{ik(a+b)}e^{Qb}-ike^{ika} \end{vmatrix}$$

$$k[e^{ik(a+b)}\sinh Qb(ike^{-ika} - Qe^{ik(a+b)}e^{Qb}) - Qe^{ik(a+b)}\cosh Qb(e^{-ika} - e^{ik(a+b)}e^{Qb})] - Q[\sin ka(-ike^{-ika} + Qe^{ik(a+b)}e^{Qb}) - k\cos ka(e^{+ika} - e^{ik(a+b)}e^{Qb})] + Q - ik[Qe^{ik(a+b)}\cosh Qb \sin ka + k\cos ka\, e^{ik(a+b)}\sinh Qb] = 0$$

$k[ike^{ik(a+b)} e^{-ika} \sinh Qb - Qe^{2ik(a+b)} \sinh Qb e^{Qb}] -$

$Qe^{ik(a+b)} \cosh Qb e^{-Qb} + Qe^{2ik(a+b)} e^{Qb} \cosh Qb -$

$Q[-ik \sin ka e^{-ika} + Q \sin ka e^{ik(a+b)} e^{Qb} - ke^{-ika} \cos ka$

$+ke^{ik(a+b)} e^{Qb} \cos ka] + Q - ik[Qe^{iQ(a+b)} \cosh Qb \sin ka$

$+k \cos ka e^{ik(a+b)} \sinh Qb] = 0$

$\Rightarrow ik^2 e^{ik(a+b)} e^{-ika} \sinh Qb - Qke^{2ik(a+b)} e^{Qb} \sinh Qb$

$-kQe^{ik(a+b)} e^{-ika} + kQe^{2ik(a+b)} e^{Qb} \cosh Qb +$

$ike^{-ika} Q \sin ka - Q^2 e^{ik(a+b)} e^{Qb} \sin ka + kQ \cos ka e^{-ika}$

$-e^{Qb} e^{ik(a+b)} kQ \cos ka + Q^2 e^{ik(a+b)} \sin ka \cosh Qb$

$+Qke^{ik(a+b)} \cos ka \sinh Qb - ikQe^{ik(a+b)} \sin ka \cosh Qb$

$-ik^2 e^{ik(a+b)} \cos ka \sinh Qb = 0$

$ik^2 e^{ik(a+b)} \sinh Qb (e^{-ika} - \cos ka) + Q^2 e^{ik(a+b)} \sin ka$

$(\cosh Qb - e^{Qb}) + kQe^{2ik(a+b)} (-\sinh Qb + \cosh Qb) e^{Qb}$

$+kQe^{ik(a+b)} \cos ka (\sinh Qb - e^{Qb}) - kQe^{ik(a+b)}$

$\cosh Qb (e^{-ika} + \sin ka) + kQe^{-ika} (\cos ka + i \sin ka) = 0$

As we know

$\because e^{\pm ika} = \cos ka \pm i \sin ka$

$\because e^{\pm Qb} = \cosh Qb \pm \sinh Qb$

$ik^2 e^{ik(a+b)} \sinh Qb (-i \sin ka) + Q^2 e^{ik(a+b)} \sin ka (-\sinh Qb)$

$+kQe^{2ik(a+b)} e^{Qb} e^{-Qb} + kQe^{ik(a+b)} \cos ka (-\cos kQb)$

$-kQe^{ik(a+b)} \cosh Qb (\cos ka) + kQe^{-ika} e^{ika} = 0$

$-i^2 k^2 e^{ik(a+b)} \sinh Qb \sin ka - Q^2 e^{ik(a+b)} \sin ka \sinh Qb$

$+kQe^{2ik(a+b)} e^{Qb} e^{-Qb} + kQe^{ik(a+b)} \cos ka \cosh Qb -$

$kQe^{ik(a+b)} \cosh Qb \cos ka + kQ = 0$

$$k^2 e^{ik(a+b)} \sin ka \sinh Qb - Q^2 e^{ik(a+b)} \sin ka \sinh Qb$$
$$+ kQ e^{2ik(a+b)} - kQ e^{ik(a+b)} \cos ka \cosh Qb - kQ e^{ik(a+b)}$$
$$\cosh Qb \cos ka + kQ = 0$$

$$(k^2 - Q^2) e^{ik(a+b)} \sin ka \sinh Qb - 2kQ e^{ik(a+b)} \cos ka \cosh Qb$$
$$+ kQ (e^{2ik(a+b)} + 1) = 0$$

Divide by $e^{ik(a+b)}$

$$(k^2 - Q^2) \sin ka \sinh Qb - 2kQ \cos ka \cosh Qb +$$
$$kQ \left(\frac{e^{2ik(a+b)}}{e^{ik(a+b)}} + \frac{1}{e^{ik(a+b)}} \right) = 0$$

$$(k^2 - Q^2) \sin ka \sinh Qb - 2kQ \cos ka \cosh Qb +$$
$$kQ (e^{ik(a+b)} + e^{-ik(a+b)}) = 0$$

$$-(Q^2 - k^2) \sin ka \sinh Qb - 2kQ \cos ka \cosh Qb$$
$$+ 2kQ \cos k(a+b) = 0$$

$$2kQ \cos k(a+b) = (Q^2 - k^2) \sin ka \sinh Qb +$$
$$2 \cos ka \cosh Qb \, kQ$$

$$\cos k(a+b) = \frac{Q^2 - k^2}{2kQ} \sin ka \sinh Qb + \cos ka \cosh Qb$$

Special case:-

By using the relation for more simplification of above eq.

$$U_0 - E = \frac{\hbar^2 Q^2}{2m}$$

$$\frac{Q^2}{2} = \frac{m}{\hbar^2} (U_0 - E)$$

Multiply B.s by "ab"

$$\frac{Q^2 ab}{2} = \frac{mu_0}{\hbar^2} ab - \frac{mE}{\hbar^2} ab$$

The potential by the periodic delta function obtained when we pass to the limit $b \to 0$ and $u_0 \to \infty$ in such a way that $u_0 b$ remain finite.

$$\lim_{\substack{u_0 \to \infty \\ b \to 0}} \left(\frac{Q^2 ab}{2} \right) = \lim_{\substack{u_0 \to \infty \\ b \to 0}} \frac{ma}{\hbar^2} (u_0 b) - \lim_{\substack{u_0 \to \infty \\ b \to 0}} \frac{E ab}{\hbar^2}$$

$$= \frac{ma}{\hbar^2} (\text{constt})$$

$$\lim_{\substack{u_0 \to \infty \\ b \to 0}} \left(\frac{Q^2 ab}{2} \right) = P = \text{constt}$$

$Q >> K$

put in eq (A).

$$\left(\frac{Q^2}{2KQ} \right) \sinh bQ \sin Ka + \cosh Qb \cos Ka = \cos K(a+b)$$

$$\because \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

$$\because \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$$

$\sinh Qb \cong Qb \quad ; \quad \cosh Qb \cong 1$

put in above eq

$$\left(\frac{Q^2}{2KQ} \right) \sin Ka \cdot Qb + \cos Ka = \cos K(a+b)$$

Multiply and divide by "a". only 1st term

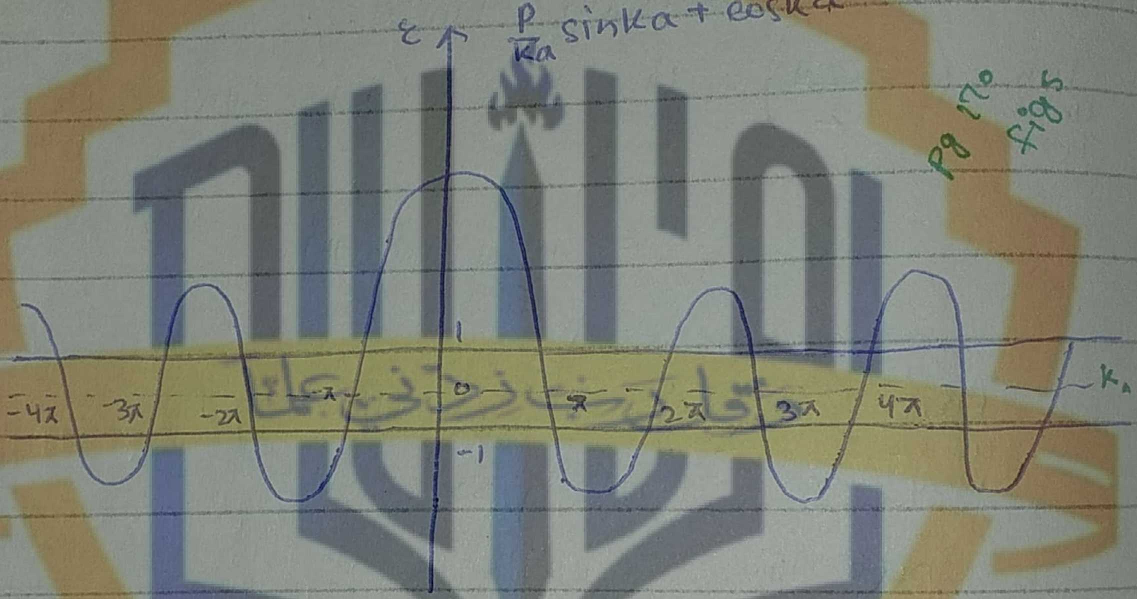
$$\left(\frac{Q^2 ab}{2Ka} \right) \sin Ka + \cos Ka = \cos K(a+b)$$

$$\because \frac{Q^2 ab}{2} = P \quad b \to 0$$

$$\frac{P}{ka} \sin ka + \cos ka = \cos k(a+0)$$

$$\cos ka = \frac{P}{ka} \sin ka + \cos ka$$

we plot a graph of this eq with ka along x-axis above eq along y-axis and take $P = \frac{3\pi}{2}$. So, $\cos ka$ lies b/w ± 1



Pg 170 Fig 5

✓ Since maximum possible value of $\cos ka$ is 1 and -1, $-1 < \cos ka < 1$.

So, the wave function of wave or solution will exist if the value of ε (energy) lies b/w ± 1

For the other value of energy there is no solution of travelling wave or Block-like solution to the wave eq.

so, there value of "ε" for which travelling wave have no solution is called forebidden energy gaps in the spectrum.

So, the energy gaps at

$ka = 0, \pi, 2\pi, 3\pi, ---)$

plot of energy vesses wave-number ka.

Fig 6
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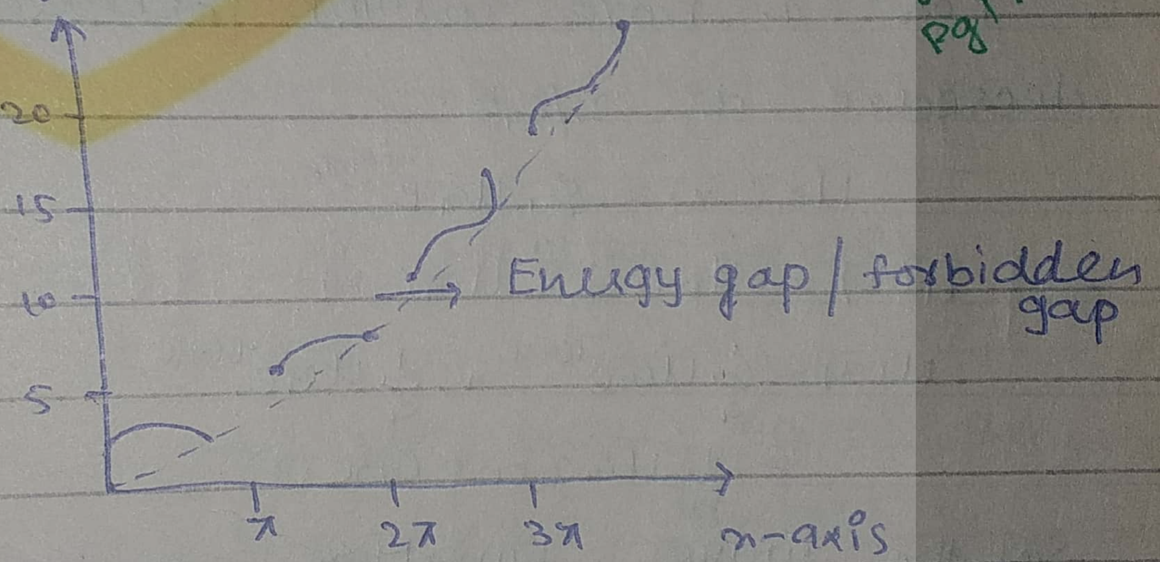


easy

EFFECT OF TEMPERATURE ON THE FERMI-DIRAC DISTRIBUTION

The ground state is the state of the N electron system at absolute zero. Q ⇒ What happens as the temperature is increased? This is a standard problem in elementary statistical mechanics, and the solution is given by the Fermi-Dirac distribution function (Appendix D and *TP*, Chapter 7).

⇒ The kinetic energy of the electron gas increases as the temperature is increased: some energy levels are occupied which were vacant at absolute zero, and some levels are vacant which were occupied at absolute zero (Fig. 3). The **Fermi-Dirac distribution** gives the probability that an orbital at energy ϵ will be occupied in an ideal electron gas in thermal equilibrium:

$$\boxed{f(\epsilon) = \frac{1}{\exp[(\epsilon - \mu)/k_BT] + 1}} \quad (5)$$

The quantity μ is a function of the temperature; ~~μ is to be chosen for the particular problem in such a way that the total number of particles in the system comes out correctly—that is, equal to N~~. At absolute zero $\mu = \epsilon_F$, because in the limit $T \to 0$ the function $f(\epsilon)$ changes discontinuously from the value 1 (filled) to the value 0 (empty) at $\epsilon = \epsilon_F = \mu$. *At all temperatures $f(\epsilon)$ is equal to $\frac{1}{2}$ when $\epsilon = \mu$*, ~~for then the denominator of (5) has the value 2.~~

for $\varepsilon < \mu$ $f(\varepsilon) = 1$ at $T = 0$
for $\varepsilon > \mu$ $f(\varepsilon) = 0$

add

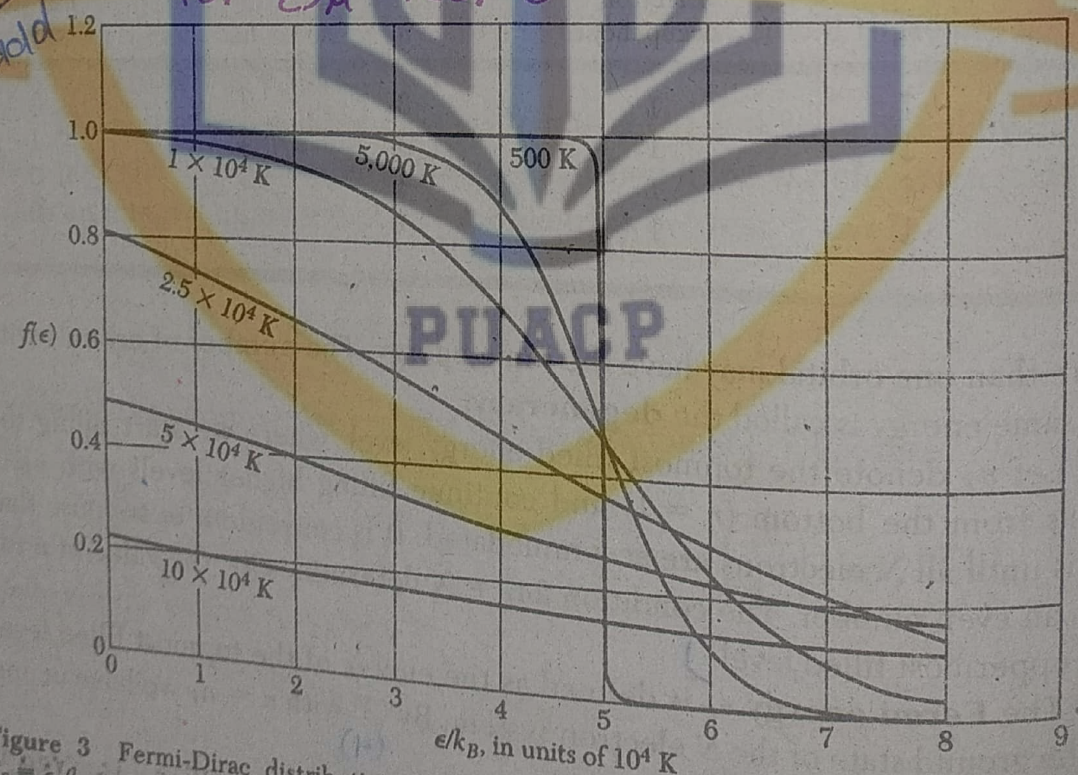


Figure 3 Fermi-Dirac distribution function (5) at the various labelled temperatures, for $T_F \equiv \epsilon_F/k_B = 50{,}000$ K. The results apply to a gas in three dimensions. The total number of particles is constant, independent of temperature. The chemical potential μ at each temperature may be read off the graph as the energy at which $f = 0.5$.

The quantity μ is the **chemical potential** (TP, Chapter 5), and we see that at absolute zero the chemical potential is equal to the Fermi energy, defined as the energy of the topmost filled orbital at absolute zero. Fig (3)

The high energy tail of the distribution is that part for which $\epsilon - \mu \gg k_B T$; here the exponential term is dominant in the denominator of (5), so that $f(\epsilon) \cong \exp[(\mu - \epsilon)/k_B T]$. This limit is called the Boltzmann or Maxwell distribution.

Q4(a) 2019:-

As we know

$$k_f = \left(\frac{3\pi^2 N}{V}\right)^{1/3}$$

$$\because \frac{N}{V} = n$$

$$k_f = \left(3\pi^2 n\right)^{1/3} \quad \text{—— (1)}$$

and

$$k_f = \frac{2\pi}{\lambda_f} \quad \text{—— (2)}$$

compare (1) and (2)

$$\frac{2\pi}{\lambda_f} = \left(3\pi^2 n\right)^{1/3}$$

$$\lambda_f = \frac{2\pi}{\left(3\pi^2 n\right)^{1/3}}$$

$$\boxed{\lambda_f = 2\left(\frac{\pi}{3n}\right)^{1/3}}$$

intrinsic

Intrinsic Carrier Concentration

As we know that at any finite temp, some electrons will acquire sufficiently thermal energy to raise from V.B to C.B. The actual number depends on the number of permissible electron energy level and the probability of these levels being occupied.

In order to calculate the intrinsic carrier concentration, we use Fermi Dirac distribution function that decides how many states are filled and how many states are empty at a certain temperature,

i.e

$$f(E) = \frac{1}{1 + e^{\frac{E - E_f}{K_B T}}}$$

The number of electrons (dn) per unit volume in the energy range E and $E + dE$ is

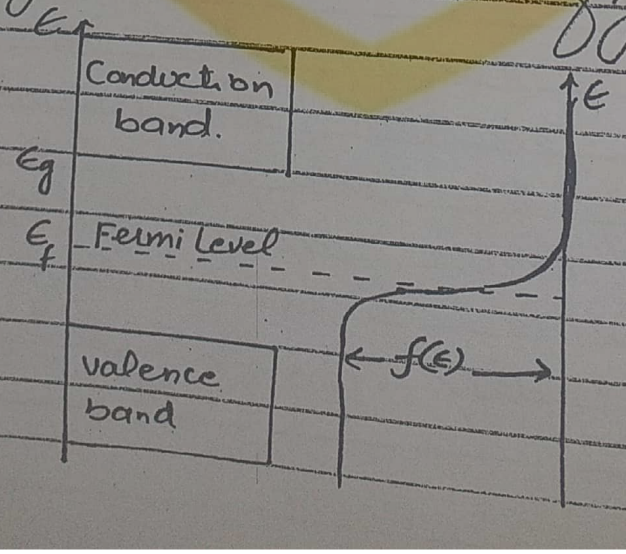
$$dn = D(E)\, f(E)\, dE$$

or

$$n = \int_{E_g}^{\infty} D(E)\, f(E)\, dE$$

where E_g is the band gap energy

Since the Fermi level for an intrinsic semi-conductor must well lie within the forbidden energy region as shown in fig



The fermi Distribution function is also shown on the same scale, for a temp $K_\beta T << E_g$. The fermi level E_f is taken to lie well within the band gap, as for an intrinsic semiconductor.

If $E = E_f$ then $f(E) = \frac{1}{2}$

we have in this case

$$E - E_f > K_\beta T$$

So we have

$$f(E) = \frac{1}{\exp\left(\frac{E - E_f}{K_\beta T}\right)}$$

or

$$f(E) = \exp\left(\frac{E_f - E}{K_\beta T}\right)$$

Density of States per unit volume is given by

$$D(E) = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(E - E_g)^{1/2}$$

Now

$$n = \int_{E_g}^{\infty} D(E) f(E)\, dE$$

$$= \int_{E_g}^{\infty} e^{\frac{E_f - E}{K_\beta T}} \cdot \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2}(E - E_g)^{1/2} dE$$

(45)

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2} e^{\epsilon_f/K_BT}\int_{\epsilon_g}^{\infty}(\epsilon-\epsilon_g)^{1/2}\, e^{-\epsilon/K_BT}\, d\epsilon$$

Now let

$$x = \frac{\epsilon-\epsilon_g}{K_BT}$$

$$\epsilon = xK_BT + \epsilon_g$$

$$d\epsilon = K_BT\,dx + 0$$

$$d\epsilon = K_BT\,dx$$

when $\epsilon = \epsilon_g$, $x = 0$

$\epsilon = \infty$, $x \to \infty$

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*}{\hbar^2}\right)^{3/2} e^{\epsilon_f/K_BT}\int_0^{\infty}(xK_BT)^{1/2}\, e^{(-xK_BT-\epsilon_g)/K_BT} \times K_BT\,dx$$

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^*K_BT}{\hbar^2}\right)^{3/2} e^{\epsilon_f/K_BT}\int_0^{\infty} x^{1/2}\, e^{-\frac{xK_BT}{K_BT}}\, e^{-\epsilon_g/K_BT}\, dx$$

$$= \frac{1}{2\pi^2}\left(\frac{2m_e^*K_BT}{\hbar^2}\right)^{3/2} e^{(\epsilon_f-\epsilon_g)/K_BT}\int_0^{\infty} x^{1/2}\, e^{-x}\, dx$$

using standard integral.

$$= \frac{1}{2\pi^2}\left(\frac{2m_e^*K_BT}{\hbar^2}\right)^{3/2} e^{\frac{\epsilon_f-\epsilon_g}{K_BT}}\ \frac{\sqrt{\pi}}{2}$$

Since $\epsilon_f - \epsilon_g = \frac{\epsilon_g}{2}$

So

$$n = \frac{1}{2\pi^2}\left(\frac{2m_e^* K_B T}{\hbar^2}\right)^{3/2} e^{\frac{\epsilon_g}{2K_BT}} \qquad \frac{\sqrt{\pi}}{2}$$

$$\boxed{n = 2\left(\frac{K_B T m_e^*}{2\hbar^2\pi}\right)^{3/2} e^{\epsilon_g/2K_BT}} \longrightarrow A$$

It is useful to calculate the equilibrium concentration of holes p. The distribution function f_h for holes is related to the electron distribution function f_e by

$$f_h = 1 - f_e$$

$$f_h = 1 - \frac{1}{e^{(\epsilon-\epsilon_f)/K_BT} + 1}$$

$$= \frac{e^{(\epsilon-\epsilon_f)/K_BT} + 1 - 1}{e^{(\epsilon-\epsilon_f)/K_BT} + 1}$$

$$f_h = \frac{e^{(\epsilon-\epsilon_f)/K_BT}}{e^{(\epsilon-\epsilon_f)/K_BT}\left[1 + e^{(\epsilon_f-\epsilon)/K_BT}\right]}$$

$$f_h = \frac{1}{e^{(\epsilon_f-\epsilon)/K_BT} + 1}$$

Since $\epsilon_f - \epsilon >> K_BT$

So $f_h = \dfrac{1}{e^{(E_f - E)/K_B T}}$

$$f_h = e^{(E - E_f)/K_B T}$$

By solving, we will get

$$\boxed{p = 2\left(\frac{m_h^* K_B T}{2\pi\hbar^2}\right)^{3/2} e^{-E_g/2K_B T}} \rightarrow B$$

p → No. of holes

Multiply A & B to obtain the equilibrium relation.

$$np = 4\left(\frac{K_B T}{2\pi\hbar^2}\right)^3 \left(m_e^* m_h^*\right)^{3/2} e^{-E_g/K_B T} \longrightarrow C$$

of the form of the law of mass action.

Remaining:

This is Carrier Concentration in C.B (of é s) or in V.B (of holes) because in intrinsic Semiconductors due to thermal excitation, és in Conduction band and holes in Valence band are equal.

NOTE:

We have nowhere assumed in derivation that material is intrinsic. the result holds in the presence of impurities as well. The only assumption made is that the distance of Fermi level from the edge of both bands should be large in Comparison with $K_B T$.

(14)

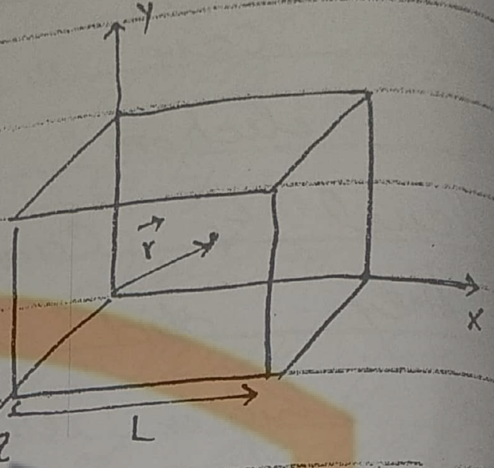
Free Electron Gas In Three Dimensions

Let us consider an electron is confined to a cube of edge L.

For this cube we have

⇒ $V(\vec{r}) = 0$ inside the box

⇒ $V(\vec{r}) \to \infty$ outside the box



⇒ The free electron S.W.E in three dimensions is written as:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\psi_k(\vec{r}) = E_k\,\psi_k(\vec{r}) \quad \text{—— (1)}$$

The solution of this wavefunction is a standing waves, that can be written as

$$\psi_k(\vec{r}) = A\sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\sin\left(\frac{n_z\pi z}{L}\right)$$

where n_x, n_y and n_z are +ve integers. → (2)

It is convenient to introduce wavefun that satisfy periodic functions, bound. The periodic boundary conditions can be written as with period 'L'

$$\psi(x+L, y, z) = \psi(x, y, z)$$
$$\psi(x, y+L, z) = \psi(x, y, z).$$
$$\psi(x, y, z+L) = \psi(x, y, z).$$

Then the Solution ② is not possible.
The Solution of travelling wave can be written as

$$\psi_K(\vec{r}) = e^{i\vec{K}\cdot\vec{r}} \longrightarrow ③$$
$$\psi_K(\vec{r}) = e^{i(K_x x + K_y y + K_z z)} \longrightarrow ④$$

Since the medium is Continuous, so the wavefunction Should satisfy the boundary Condition

i.e $\psi(x=0) = \psi(x=L)$

$$e^{iK(0)} = e^{iKL}$$
$$1 = \cos KL + i\sin KL$$

or $1 + 0i = \cos KL + i\sin KL$

Hence we have

$$K_x = 0, \frac{2\pi}{L}, \frac{4\pi}{L}, \frac{6\pi}{L}, \ldots\ldots = \frac{2\pi n_x}{L}$$

where $n_x = 0, \pm 1, \pm 2, \pm 3, \ldots\ldots$

Similarly

$$K_y = \frac{2\pi n_y}{L}$$

and $K_z = \frac{2\pi n_z}{L}$

(6)

The eq (4) should satisfy the ~~boundary~~ S.W.E. For simplicity, we consider only x-component

$$x = x + L$$

$$\psi_{K_x}(x+L) = e^{iK_x(x+L)}$$

$$= e^{i\left(\frac{2\pi n_x}{L}\right)(x+L)}$$

$$= e^{i\left(\frac{2\pi n_x x}{L}\right)} \cdot e^{i(2\pi n_x)}$$

$$= e^{i\left(\frac{2\pi n_x x}{L}\right)} \cdot \left[\cos 2\pi n_x + i \sin 2\pi n_x\right]$$

$$\psi_{K_x}(x+L) = e^{iK_x x} \cdot 1$$

$$\boxed{\psi_{K_x}(x+L) = \psi_{K_x}(x)}$$ Satisfied.

ENERGY OF ORBITAL

As we have

$$\psi_K(\vec{r}) = e^{i(\vec{K}\cdot\vec{r})}$$

$$\psi_K(\vec{r}) = e^{i(K_x x + K_y y + K_z z)}$$

$$\frac{\partial \psi_K(\vec{r})}{\partial x} = iK_x e^{i(\vec{K}\cdot\vec{r})}$$

$$\frac{\partial^2 \psi_k(\vec{r})}{\partial x^2} = -K_x^2 \, e^{i(\vec{K}\cdot\vec{r})} \rightarrow (a)$$

Similarly

$$\frac{\partial^2 \psi_k(\vec{r})}{\partial y^2} = -K_y^2 \, e^{i(\vec{K}\cdot\vec{r})} \rightarrow (b)$$

$$\frac{\partial^2 \psi_k(\vec{r})}{\partial z^2} = -K_z^2 \, e^{i(\vec{K}\cdot\vec{r})} \rightarrow (c)$$

By Adding (a), (b), (c)

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_k(\vec{r}) = -(K_x^2 + K_y^2 + K_z^2)\, e^{i(\vec{K}\cdot\vec{r})}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_k(\vec{r}) = -(K_x^2 + K_y^2 + K_z^2)\,\psi_k(\vec{r})$$

Multiplying both sides by $\frac{-\hbar^2}{2m}$

$$\frac{-\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_k(\vec{r}) = \frac{\hbar^2}{2m}(K_x^2 + K_y^2 + K_z^2)\,\psi_k(\vec{r})$$

$$\frac{-\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi_k(\vec{r}) = \frac{\hbar^2 K^2}{2m}\,\psi_k(\vec{r}) \rightarrow (5)$$

By Comparing (1) & (5), we have

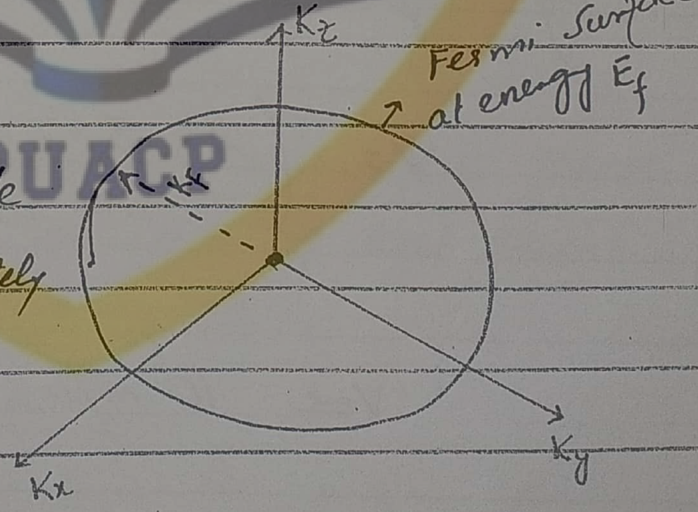
$$\boxed{E_k = \frac{\hbar^2 K^2}{2m}} \quad \text{where} \quad K = \frac{2\pi}{\lambda}$$

Fermi Sphere

In the ground state of a system of 'N' free electrons, the occupied orbitals may be represented by points inside the sphere in K-space. This sphere is known as Fermi Sphere and its surface is known as Fermi Surface. The energy of the electrons at the surface of sphere is called Fermi energy i.e maximum energy of electrons lie on the surface of sphere and is given by

$$E_f = \frac{\hbar^2 K_f^2}{2m}$$

where K_f (Fermi wave vector) is the radius of Fermi Sphere.



All the points inside the sphere are completely filled and outside are empty because they correspond to energy greater than fermi energy E_f which are occupied at 0K.

THE DENSITY OF STATES:

In the ground state of a system of 'N' free electrons the occupied orbitals may be represented as points inside a sphere in K-space. The energy at the surface of the sphere is the Fermi Energy; the wavevectors at the Fermi surface have a magnitude K_f such that

$$E_f = \frac{\hbar^2 K_f^2}{2m} \longrightarrow (1)$$

The volume occupied by each point may be represented as $\left(\frac{2\pi}{L}\right)^3$ of K-space and volume of whole sphere is $\frac{4}{3}\pi K_f^3$. Then total no of orbitals is

$$n = \frac{N}{2}$$

where

$$n = \frac{\text{volume occupied by whole sphere}}{\text{volume occupied by each point}}$$

$$\frac{N}{2} = \frac{4/3\, \pi K_f^3}{(2\pi/L)^3}$$

$$N = \frac{2 \times 4/3\, \pi K_f^3}{8\pi^3/L^3} = \frac{L^3 \pi K_f^3}{3\pi^3}$$

$$N = \frac{V}{3\pi^2} K_f^3$$

$$\boxed{K_f = \left(\frac{3\pi^2 N}{V}\right)^{1/3}} \longrightarrow (2)$$

Which depends only on the particle Concentration and not on the mass.

From eq (1)

$$E_f = \frac{\hbar^2 K_f^2}{2m}$$

By using (2)

$$E_f = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \longrightarrow (3)$$

This relates the Fermi Energy to the electron Concentration $\frac{N}{V}$. The electron velocity at the Fermi Surface is

$$V_f = \frac{\hbar K_f}{m}$$

using (2)

$$\boxed{V_f = \frac{\hbar}{m}\left(\frac{3\pi^2 N}{V}\right)^{1/3}} \longrightarrow (3)$$

⇒ we now find and expression for the number of orbitals per unit energy range $D(E)$, often called the density of states) We use eq ② for the total no. of orbitals of energy $\leq E_f$

$$E_f = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3}$$

$$\frac{2mE_f}{\hbar^2} = \left(\frac{3\pi^2 N}{V}\right)^{2/3}$$

$$\frac{3\pi^2 N}{V} = \left(\frac{2mE_f}{\hbar^2}\right)^{3/2}$$

$$N = \frac{V}{3\pi^2}\left(\frac{2mE_f}{\hbar^2}\right)^{3/2} \quad ⑤$$

The density of orbitals can be defined as

$$D(E) = \frac{dN}{dE_f}$$

$$= \frac{V}{3\pi^2} \times \frac{3}{2}\left(\frac{2mE_f}{\hbar^2}\right)^{1/2} \frac{2m}{\hbar^2}$$

$$\boxed{D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_f^{1/2}} \rightarrow ⑥$$

$$= \frac{3\varepsilon_f}{3\varepsilon_f}\frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_f^{1/2}$$

From this equation we can say

$$D(E) \propto E_f^{1/2}$$

$$= \frac{3}{2\varepsilon_f}\left[\frac{V}{3\pi^2}\left(\frac{2m}{\hbar^2}E_f\right)^{3/2}\right]$$

The graph can be plotted as

$$\boxed{D(\varepsilon) = \frac{3N}{2\varepsilon_f}}$$

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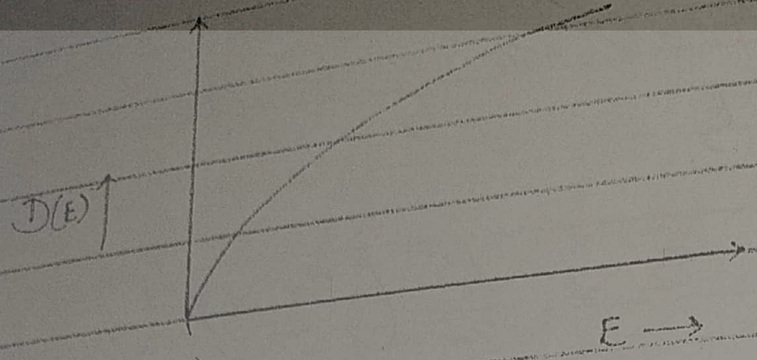

The graph shows that as energy increases the no of states increases i.e Density of states tells us that how many states of an energy are present.

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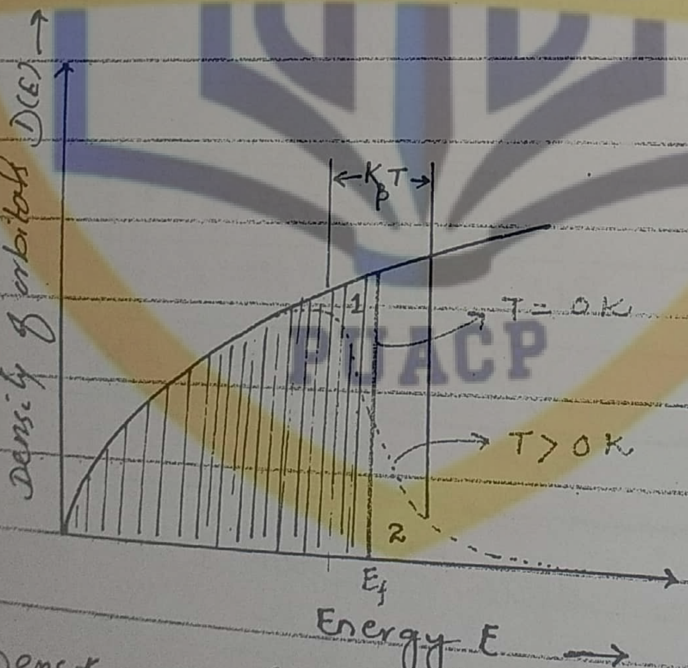
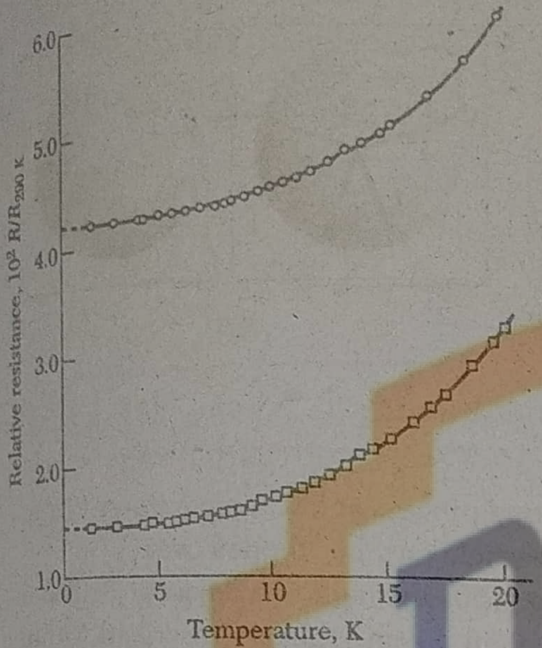


Fig: Density of Single particle States as a function of energy, for a free electron gas in



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(atom, molecule a phonon)

⇒ Mean free path is the average distance travelled by a moving particle b/w successive impacts which modifies its direction or energy or other particle properties

Figure 12 Resistance of potassium below 20 K, as measured on two specimens by D. MacDonald and K. Mendelssohn. The different intercepts at 0 K are attributed to different concentrations of impurities and static imperfections in the two specimens.

impurity concentration of about 20 ppm. In exceptionally pure specimens the resistivity ratio may be as high as 10^6, whereas in some alloys (e.g., manganin) it is as low as 1.1.

It is possible to obtain crystals of copper so pure that their conductivity at liquid helium temperatures (4 K) is nearly 10^5 times that at room temperature; for these conditions $\tau \approx 2 \times 10^{-9}$ s at 4 K. The mean free path ℓ of a conduction electron is defined as

$S = vt$

$$\ell = v_F \tau \ , \tag{47}$$

where v_F is the velocity at the Fermi surface, because all collisions involve only electrons near the Fermi surface. ~~From Table 1 we have~~ $v_F = 1.57 \times 10^8$ cm s^{-1} for Cu) thus the mean free path is $\ell(4\text{ K}) \approx 0.3$ cm. Mean free paths as long as 10 cm have been observed in very pure metals in the liquid helium temperature range. This mean free path for copper is

$\ell = v_F \tau$

$= (1.57 \times 10^8)(2 \times 10^{-9} s)$

$\ell(4K) = 3.14 \times 10^{-1}$ cm

$\ell = 0.3$ cm.

The temperature-dependent part of the electrical resistivity is proportional to the rate at which an electron collides with thermal phonons and thermal electrons. The collision rate with phonons is proportional to the concentration of thermal phonons. One simple limit is at temperatures over the Debye temperature θ: here the phonon concentration is proportional to the temperature T, so that $\rho \propto T$ for $T > \theta$. A sketch of the theory is given in Appendix J.

Umklapp Scattering

Umklapp scattering of electrons by phonons (Chapter 5) accounts for most of th

The electrical resistivity ρ is defined as th
so that

$$\rho = m/ne^2\tau \ . \qquad (4$$

Values of the electrical conductivity and resistivity of the elements are given i
Table 3. In Gaussian units σ has the dimensions of frequency.

It is easy to understand the result (43) for the conductivity of a Fermi ga
We expect the charge transported to be proportional to the charge density n
the factor e/m enters (43) because the acceleration in a given electric field
proportional to e and inversely proportional to the mass m. The time τ describ
the free time during which the field acts on the carrier. Closely the same resu
for the electrical conductivity is obtained for a classical (Maxwellian) gas of ele
trons, as realized at low carrier concentration in many semiconductor problems

collision time

Experimental Electrical Resistivity of Metals

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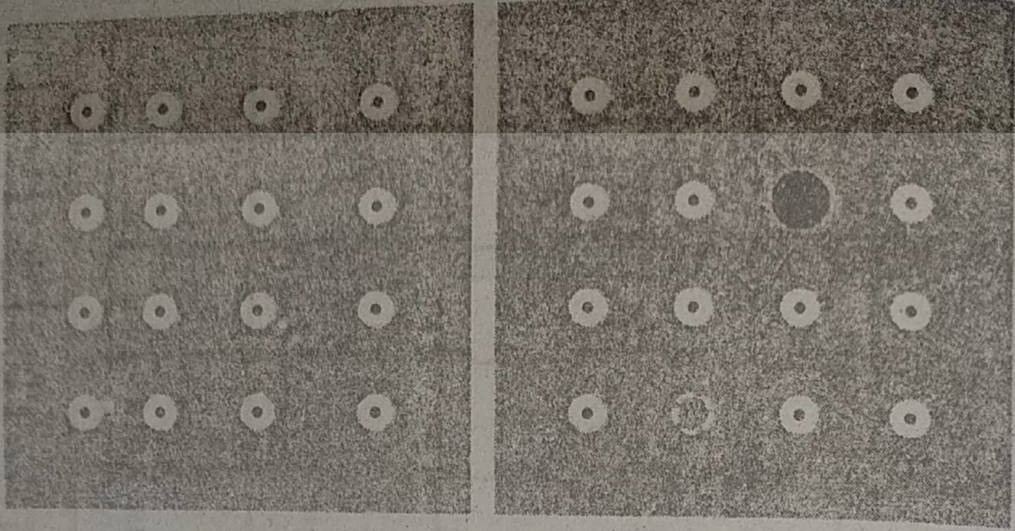
The electrical resistivity of most metals is dominated at room temperatur
(300 K) by collisions of the conduction electrons with lattice phonons and a



(a) (b)

Figure 11 Electrical resistivity in most metals arises from collisions of electrons with irregularities in the lattice, as in (a) by phonons and in (b) by impurities and vacant lattice sites.

liquid helium temperature (4 K) by collisions with impurity atoms and mechanical imperfections in the lattice (Fig. 11). The rates of these collisions are often independent to a good approximation, so that if the electric field were switched off the momentum distribution would relax back to its ground state with the net relaxation rate

$$\frac{1}{\tau} = \frac{1}{\tau_L} + \frac{1}{\tau_i} , \tag{45}$$

where τ_L and τ_i are the collision times for scattering by phonons and by imperfections, respectively.

The net resistivity is given by

$$\rho = \rho_L + \rho_i , \tag{46}$$

where ρ_L is the resistivity caused by the thermal phonons, and ρ_i is the resistivity caused by scattering of the electron waves by static defects that disturb the periodicity of the lattice. Often ρ_L is independent of the number of defects when their concentration is small, and often ρ_i is independent of temperature. This empirical observation expresses **Matthiessen's rule**, which is convenient in analyzing experimental data (Fig. 12). this become less accurate at high temp and high impurity content

The residual resistivity, $\rho_i(0)$, is the extrapolated resistivity at 0 K because ρ_L vanishes as $T \to 0$. The lattice resistivity, $\rho_L(T) = \rho - \rho_i(0)$, is the same for different specimens of a metal, even though $\rho_i(0)$ may itself vary widely. The **resistivity ratio** of a specimen is usually defined as the ratio of its resistivity at room temperature to its residual resistivity. It is a convenient approximate indicator of sample purity: for many materials an impurity in solid solution creates a residual resistivity of about 1 μohm-cm (1×10^{-6} ohm-cm) per atomic percent of impurity. A copper specimen with a resistivity ratio of 1000 will have a residual resistivity of 1.7×10^{-3} μohm-cm, corresponding to an